

The design of this database fully prioritizes the data integrity and operational for a mobile money system. This is by isolating base identities in the users table from financial balances in the wallet table through a strict one-to-one relationship and safeguards balance data while keeping user profile management. Rather than duplicating identity credentials for specialized actors, the agents table links back to users in a one-to-optional relationship, allowing standard platform users to seamlessly operate as cash kiosks without redundant records.

To handle core financial operations efficiently, the transactions entity uses a unified structure that eliminates the need for separate tables across different transaction types. Linking transactions back to users via two distinct foreign keys— `sender_id` and `receiver_id` —allows the transfers to be captured in a single, clean database record. Furthermore, including an optional `agent_id` foreign key enables cash-in deposits and cash-out withdrawals to leverage the exact same schema.

Finally, security and fault tolerance are reinforced through the `transaction_logs` table, which maintains a one-to-many relationship with transactions. Decoupling status transitions from live transaction processing ensures an immutable audit trail. Every state shift from pending to completed or failed is recorded independently, providing the historical tracking necessary for fraud detection, system recovery, and dispute resolution without impacting active transaction performance.